

HIGH-END CHIPSETS: BLEEDING-EDGE REALISM REDEFINED!

Thanks to the hype DX 10 created, we've seen a lot more pixel-crunching power on the new breed of cards. Both NVIDIA's 8800 and ATI's HD2900 series GPUs are much faster than their predecessors—thanks to an insane jump in the number of shader units.

These cards just chew up the latest DX 9 games and swallow them without a hiccup! Gone are the days when AntiAliasing used to cause stuttering in frames—the 8800GTX and HD2900XT will handle Anisotropic Filtering and AntiAliasing with a hit of anywhere between 5 and 15 frames even at high resolutions. This bespeaks tremendous improvements in the overall architecture and FSAA algorithms. Needless to say, both these cards will handle all the upcoming titles with ease. And you won't need to tone down settings for some time to come.

The Contenders

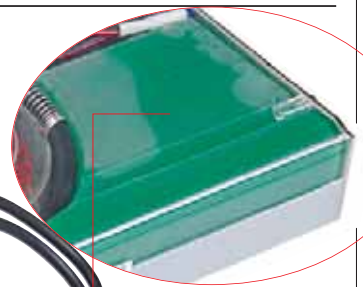
As they say, *it's lonely at the top*. ATI's HD2900XT is still a very new chipset, and we were only able to get a single card from one vendor—MSI. The MSI RX2900XT VT2D512-E comes with all the connectors you'll likely need. A DVI to HDMI adapter is provided—very important for HD junkies who will likely pair this card with a large screen LCD or Plasma screen. Incidentally,

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**Asus EN8800GTX
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shader units. It's also a shorter card—the PCB (Printed Circuit Board) being 1.5 inches shorter than the PCB of the 8800GTX/Ultra cards. Most of the vendors are generous, and the games bundled are some of the newer titles—especial-



The water reservoir



The heatpipes

Parameters	8800 Ultra	8800GTS	HD2900XT	X1950XTX	X1900XTX
No. Of Transistors (million)	681	681	700	384	384
Core Speed (MHz)	612	500	740	650	650
Number of Shader Units	128	96	320	48(PS)+8(VS)*	48(PS)+8(VS)*
Memory Bandwidth (GBps)	103.2	64	106	51.2	48

*PS= Pixel Shader, VS= Vertex Shader

the HD2900XT chipset also has an audio solution integrated (HDMI supports video and audio signals on a single connect). There's also a token bundled that'll get you three upcoming Valve Software game titles, when they release of course... This card is also phenomenally heavy; we estimate it to be at least 200 grams more than the 8800GTX/Ultra (the next-heaviest card). This is because of the generous use of copper in the cooling solution, and three copper heat pipes.

NVIDIA had a slightly better showing with one 8800 Ultra from Galaxy and three 8800 GTX cards—two from ASUS and one from XFX. ASUS' 8800GTX Aquatank was something special—it features a liquid cooling solution and is completely set up to go with water block, pump and radiator. Since liquid cooling is thermally more efficient, ASUS overclocked this card to beyond 8800 Ultra levels. The radiator is well designed and looks like any other card—it'll occupy an extra PCI slot. The other three cards used reference NVIDIA coolers.

There were seven GeForce 8800GTS-based cards in our test—all on reference coolers. Note that the 8800GTS is a lower version of the 8800GTX architecture with slower core and memory speeds, less memory bus-width and

